

Fraud Detection

Identifying potential fraudulent transactions and enhancing fraud detection capabilities with XGBoost for ASB



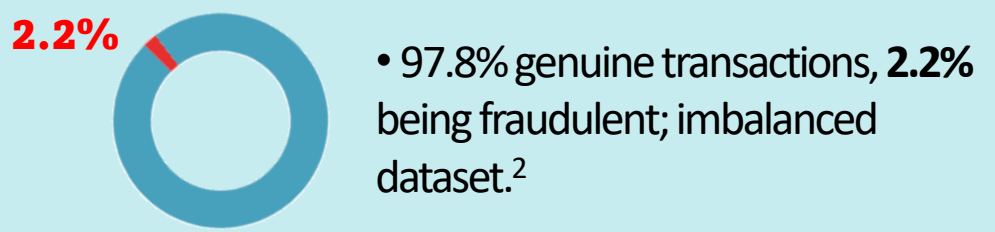
INTRODUCTION

Background

• To prevent erosion of customer confidence and mitigate significant monetary losses—which have increased 20-fold from 2020 to 2023—detecting and responding to fraudulent transactions has become an urgent challenge for ASB that must be addressed.¹

Dataset Overview

• **500,000** synthetically generated transactions from 10,000 customers from ASB.



METHODS

Data Cleaning

- **Transaction Hour Extracted**
- **Agents vs Transaction Types**
ATM, Mobile Device, Web Browser, Others
- **Factor Numerical Variables**
FraudLabel, Transaction Amount in Five Intervals

Techniques

✓ Selected Predictors

Age, Hour, TransactionType, TransactionAmount



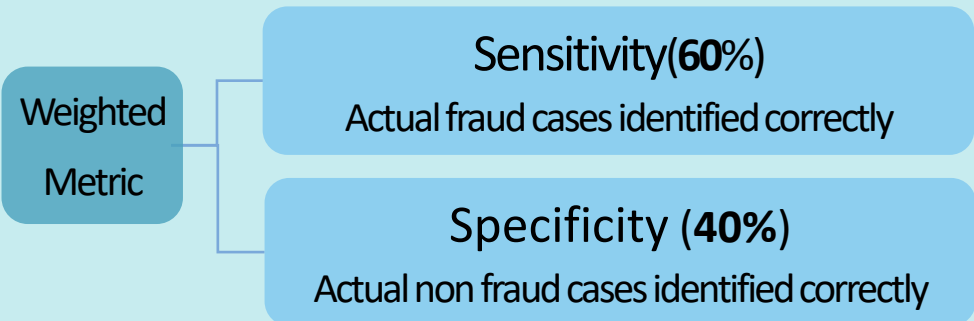
✓ Address class imbalance SMOTE

Synthetic Minority Over-sampling Technique



✓ Employed Models

LightGBM, Logistic Regression, and XGBoost*.



Assumption

All the predictors are highly relevant and independent

* XGBoost, which is chosen later, is a machine learning algorithm that uses gradient boosting to build an ensemble of decision trees, optimizing model performance through iterative corrections and regularization to enhance predictive accuracy and prevent overfitting.

REFERENCES

1. Ministry of Business, Innovation and Employment. (2023). 198 million dollars lost to scams in the last year. Retrieved August 13, 2024, from <https://www.mbie.govt.nz/about/news/198-million-dollars-lost-to-scams-in-the-last-year>
2. Zabój, M., et al. (2021). Comprehensive fraud detection and verification process. KPMG. Retrieved from <https://kpmg.com/pl/en/home/insights/2021/02/comprehensive-fraud-detection-and-verification-process.html>

RESULTS

Model Performance

XGBoost outperformed other models, achieving the highest specificity (0.797) with a high accuracy of 0.8, which minimizes consumer monetary loss by accurately catching true positives and improves bank operational efficiency by ensuring customers are not unnecessarily bothered. It is highly effective in distinguishing between "fraud" and "non fraud" transactions while obtaining the highest weighted metrics (0.72).



Figure 1. Prediction Classification Result Summary (Sample = 125,000 transactions)

Key predictor: Excluding age, transaction types *Withdrawal* contributes to most of the model's predictive power. Other transaction types and transaction amount ranges come after.

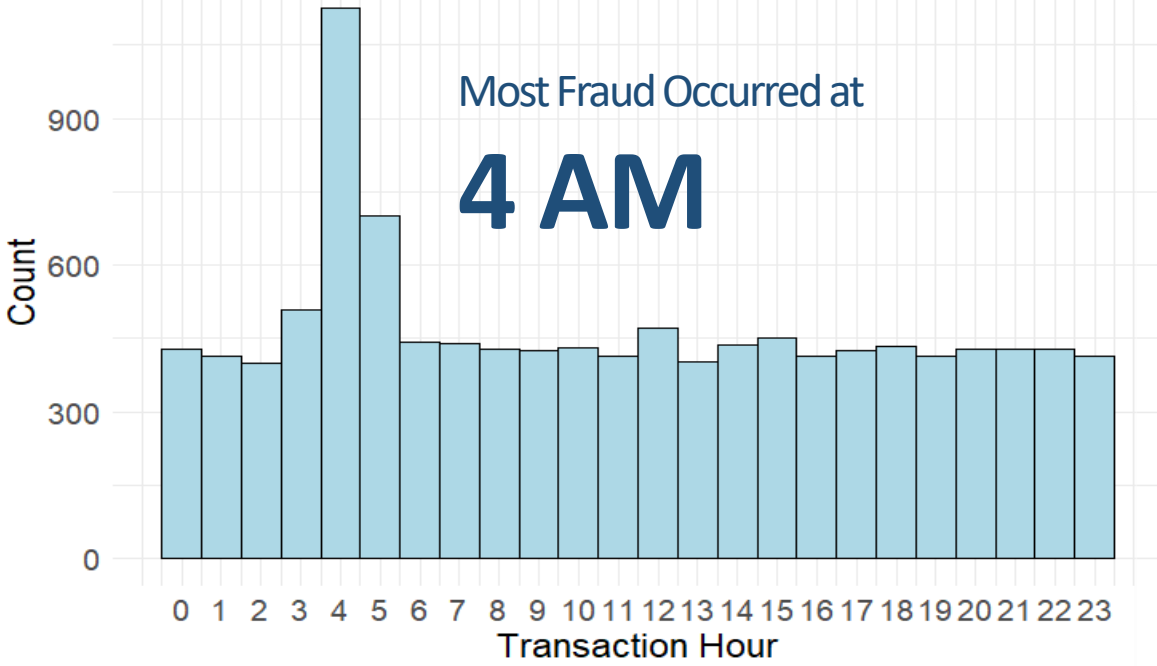


Figure 3. Distribution of Fraud by Transaction Hour

Older ages are more susceptible to fraudulent transactions.

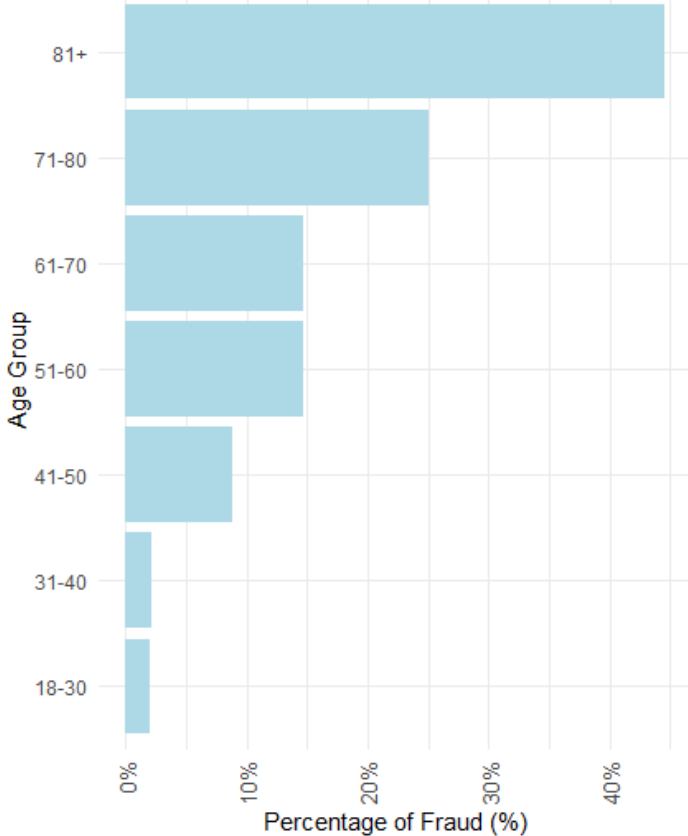


Figure 2. Percentage of Individuals Experienced Fraud by Age Group

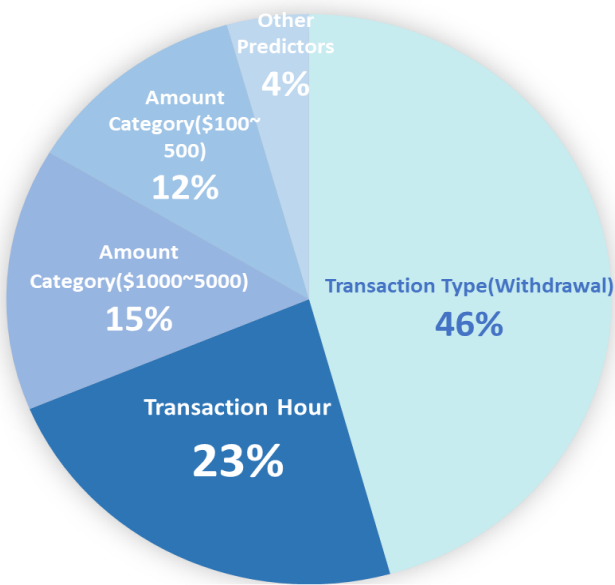


Figure 4. Predictors Importance Excluded Age

CONCLUSION

Powered by XGBoost and enhanced by SMOTE sampling, this model effectively identifies fraudulent transactions from May to July for ASB. The performance is measured using a weighted metric that balances sensitivity and specificity, ensuring effective fraud detection while minimizing false alarms that could inconvenience customers.

RECOMMENDATION

• **Introducing two-factor authentication (2FA) to enhance the security process:** requires text confirmation during 2-6 AM as the second factor to grant access to ATM withdraw. This strategy reduces the risk of unauthorized withdraw while maintains low cost and customer impact.



• **Internal monitoring and mass mailing customers to make customers aware of the latest fraud trends:** Keeping regular internal monitoring of irregular transactions and alerting customers through push notifications of the latest trends in fraudulent transactions. This strategy will improve customer awareness and is a low cost solution for reducing potential fraudulent transactions.



• **Customers can customize their transactions limit per day, per account, per transaction type.** There is a higher probability of fraudulent transitions in transaction amount \$1000-5000 (relatively high transaction amount). However, the transaction amount is a predictor but not a strong indicator.

